

Predictors of the study event in a dietary intervention study on fasting glucose

Abstract

Background: A dietary intervention study on fasting glucose. Methods: The odds of the study event were modelled with a maximum-likelihood logistic regression (no penalisation) with the Mediterranean diet intervention (coded 1 for mediterranean, 0 for usual), age and body-mass index as predictors. Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

Introduction

We report a dietary intervention study on fasting glucose. The primary question was whether the Mediterranean diet intervention is associated with fasting glucose (mg/dL) at follow-up. The dataset accompanying this report (data.csv) contains one row per participant and is described in the data dictionary.

Methods

Participants and data. Data were collected for 193 participants. Each record contains the variables listed in the data dictionary. Group membership is recorded in the column `diet`.

Exclusion criteria and preprocessing. Only participants aged 18 to 80 years inclusive were included in the analysis. Participants with missing values for glucose_fu, bmi were excluded (complete-case analysis).

Statistical analysis. The odds of the study event were modelled with a maximum-likelihood logistic regression (no penalisation) with the Mediterranean diet intervention (coded 1 for mediterranean, 0 for usual), age and body-mass index as predictors. Odds ratios with Wald 95% confidence intervals are reported. All analyses were performed on the analysis sample defined above. Statistics were computed with standard scientific software; no random resampling was used.

Results

Among 161 participants there were 59 events. The mediterranean diet intervention was associated with an adjusted odds ratio of 0.35 (95% CI 0.16 to 0.73, $p = 0.005$); the odds ratio per year of age was 1.02.

Discussion

The analysis followed the pre-specified plan described in the Methods. Limitations include the observational nature of some comparisons and the exclusion of records with missing values.